Problem Sheet 3

discussion: Tuesday, 27.10.2026

3.1. Let I_n be the Lagrangian interpolation operator of degree n in distinct knots $x_0, \dots, x_n \in [-1, 1]$ with Lagrange polynomials $\ell_0, \dots, \ell_n$. Show that:

a) The mapping $f \mapsto I_n f$ is a linear map, i.e., for continuous functions f, g and $\lambda \in \mathbb{R}$ there hold $I_n(f + g) = (I_n f) + (I_n g)$ as well as $I_n(\lambda f) = \lambda I_n f$.

b) $I_n f = f$ for all polynomials $f \in \mathcal{P}_n$.

c) For perturbed values $\widetilde{f}_i$ with $|\widetilde{f}_i - f(x_i)| \le \delta$, the "perturbed" interpolation polynomial $\sum_i \widetilde{f}_i \ell_i$ satisfies

$$\|(\sum_{i=0}^{n} \widetilde{f}_i \ell_i) - I_n f\|_{\infty,[-1,1]} \le \Lambda_n \delta,$$

where Λ_n denotes the Lebesgue constant.

3.2. Let the data $(0, 0), (1, 1), (2, 1), (3, 2), (4, 2)$ be given.

a) How many degrees of freedom does the cubic spline with a not-a-knot condition have? Write down the linear system of equations that has to be solved to compute this cubic spline!

b) Solve the linear system (not necessarily by hand!) and write down and draw the cubic spline as a function.

c) Assume that the given values $f_i = f(x_i)$ above are evaluations of a given function f with $\|f^{(4)}\|_{\infty,[0,4]} \le 4$. Write down an upper bound for the error

$$\|f - p\|_{\infty,[0,4]},$$

where p denotes the cubic spline.

3.3. For $n \in \{10, 20, 40\}$ consider interpolation on the interval $[-5, 5]$ in $n+1$ points. Compare the uniformly distributed points $x_i^{unif} := 5(-1 + 2\frac{i}{n})$, $i = 0, \dots, n$ and the Chebyshev points $x_i^{Cheb} = 5\cos(\frac{i+0.5}{n+1}\pi)$, $i = 0, \dots, n$.

a) For the function $f(x) = (1 + x^2)^{-1}$, plot the two interpolating polynomials on $[-5, 5]$.

b) Investigate numerically the Lebesgue constant

$$\Lambda_n := \max_{x \in [-5,5]} \sum_{i=0}^{n} |\ell_i(x)|,$$

for both the uniform interpolation point distribution and the Chebyshev points. To that end, plot Λ_n versus n in semilogarithmic scale (`semilogy`).

c) For the uniform point distribution, one can in fact show that $\Lambda_n \approx Ce^{bn}$ for some C, $b > 0$. Determine C and b from your data as follows: take the logarithm $\log \Lambda_n \approx \log C + bn$ and fit this line to your data for $n \in \{20, 40\}$. *Hint:* you can let `polyfit` compute $\log C$ and b, or solve a 2×2 system.

3.4. Consider the oscillatory function

$$f(x) = \cos(\pi x) + \frac{1}{4}\cos(3\pi x), \qquad x \in [-1, 1],$$

and the knots $x_0 = -1$, $x_1 = 0$, $x_2 = 1$.

a) Compute the natural cubic spline s_{nat} interpolating f at the given knots, with $s''_{nat}(-1) = s''_{nat}(1) = 0$. Write down its polynomial pieces and sketch f and s_{nat}. Also determine the complete/clamped cubic spline s_{cl} with $s'_{cl}(-1) = f'(-1)$ and $s'_{cl}(1) = f'(1)$.

b) Determine the errors

$$\|f - s_{nat}\|_{\infty,[-1,1]}, \qquad \|f - s_{cl}\|_{\infty,[-1,1]}.$$

You may locate the extrema numerically. Which spline has the smaller maximum error? Which of the two boundary conditions is covered by Theorem 1.32? Compute $f''(-1)$ and $f''(1)$ and compare with the natural boundary conditions.

c) Now consider a localized peak with an oscillatory background,

$$g(x) = \exp\bigl(-100(x - \tfrac{1}{5})^2\bigr) + \frac{1}{5}\sin(6\pi x), \qquad x \in [0,1].$$

For $n \in \{8, 16, 32, 64, 128, 256\}$, compute the natural and the complete/clamped cubic interpolating splines in the knots $x_j = j/n$, $j = 0, \ldots, n$. For the clamped spline, use the exact endpoint derivatives. You may use built-in spline routines in `matlab` or `python`, but specify the boundary conditions explicitly.

Plot g and both splines for $n = 8$ and $n = 32$. Approximate both maximum errors by sampling at 10001 uniformly distributed points in $[0, 1]$. Plot the errors versus $h = 1/n$ in a `loglog` plot and include the auxiliary lines $h \mapsto h^2$ and $h \mapsto h^4$. What convergence rates do you observe? How do the endpoint conditions help explain the difference?

3.5. Let $n = 5$, $x_j = 2\pi j/5$, $j = 0, \ldots, 4$, and let

$$f(x) = 1 + 2\cos(x) + \sin(2x), \qquad y_j = f(x_j).$$

As in the lecture notes, set $\omega_5 = \exp(-2\pi\mathbf{i}/5)$, where $\mathbf{i}^2 = -1$, and $V_5 = (\omega_5^{jk})_{j,k=0}^{4}$.

a) Use the identity

$$\sum_{j=0}^{4} \omega_5^{jr} = \begin{cases} 5, & r \in 5\mathbb{Z}, \\ 0, & r \notin 5\mathbb{Z}, \end{cases} \qquad r \in \mathbb{Z},$$

to derive the coefficient formula $\mathbf{c} = \frac{1}{5}V_5\mathbf{y}$ for the modified trigonometric interpolant

$$p(x) = \sum_{k=0}^{4} c_k e^{\mathbf{i}kx}.$$

Compute the coefficients c_k and write down p explicitly.

b) Replace the frequencies 3 and 4 in p by -2 and -1, respectively, to obtain a classical trigonometric polynomial

$$q(x) = a_0 + a_1\cos(x) + b_1\sin(x) + a_2\cos(2x) + b_2\sin(2x).$$

Why does q interpolate the same data? Determine its real coefficients and compare q with f. Evaluate $p(\pi)$ and $q(\pi)$. Must the modified interpolant of real data be real-valued between the knots?

c) Show that, for all integers k, ℓ,

$$e^{\mathbf{i}(k+5\ell)x_j} = e^{\mathbf{i}kx_j}, \qquad j = 0, \ldots, 4.$$

Use this identity to determine the classical trigonometric interpolant of degree at most 2 for $g(x) = f(x) + \cos(6x)$ at the same knots. Explain why the sampled data cannot distinguish the frequencies 1 and 6 (*aliasing*).